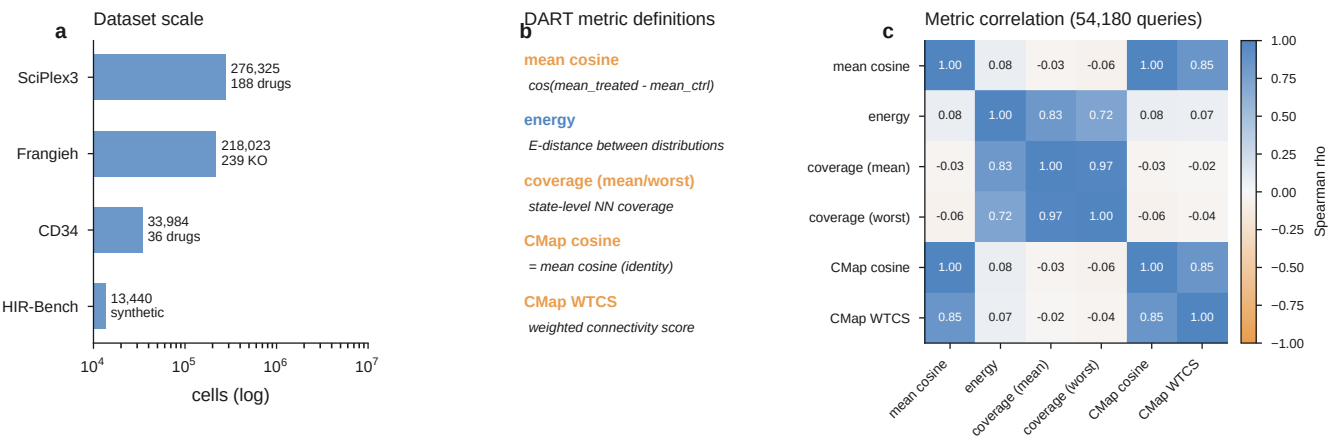


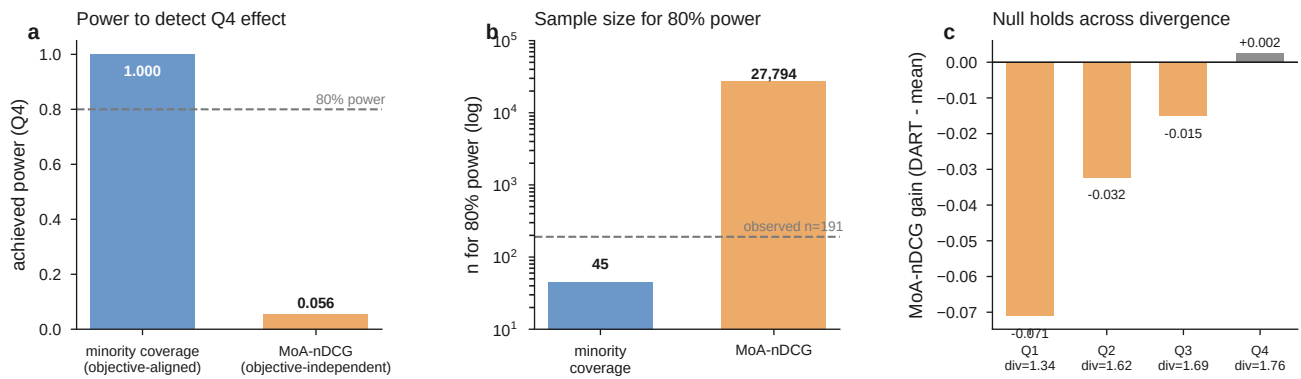
# Supplementary Information: Objective-aligned evaluation inflates the distributional gain in single-cell drug retrieval

[[Author list, as in the main manuscript]]

This Supplementary Information contains seven Extended Data figures, three Supplementary Notes, and four Supplementary Tables.

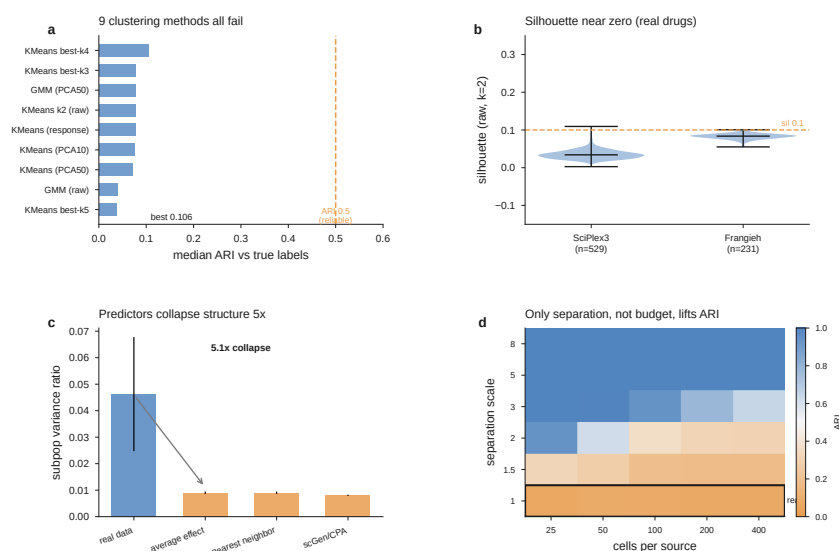


**Extended Data Fig. 1. Reproducibility basis.** **a**, Dataset scale across the four benchmarks (cells, perturbations). **b**, Distributional metric-family definitions. **c**, Spearman correlation among all retrieval metrics, computed over the 54,180 query–candidate scores generated by 1,260 retrieval queries (43 candidates each; the unit is the scored pair, not the query): mean cosine and CMap cosine are perfectly correlated ( $\rho = 1.00$ ), while energy and coverage metrics form a separate, near-orthogonal block.

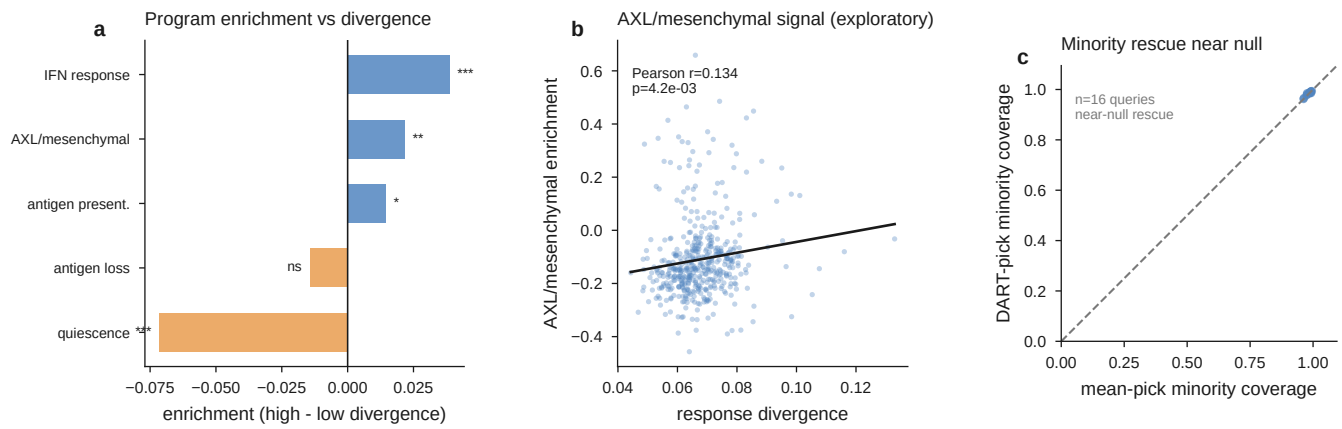


gain stays  $\leq 0$  across all divergence strata (not a power artifact)

**Extended Data Fig. 2. The Class B negative is not a power artifact.** **a**, Achieved power to detect the Q4 effect: the task-proximal minority-coverage metric is fully powered (1.00,  $n=191$ ), the task-proximal MoA-nDCG is not (0.06,  $n=143$ ). **b**, Sample size for 80% power: 45 versus 20,844 queries. **c**, MoA-nDCG gain by divergence stratum. It is significantly *negative* in Q1 (median  $-0.030$ , Benjamini-Hochberg  $q = 5.8 \times 10^{-5}$ ) and has a median of exactly zero in Q2–Q4, with no stratum showing a significant positive gain; the highest-divergence stratum’s small positive *mean* ( $+0.003$ ) is not significant ( $q = 0.57$ ). MoA-nDCG is undefined for the leave-MoA-out and partial-library splits, where the mechanism class is removed from the library, so those 165 of 765 queries are excluded and  $n=600$ . A power deficit alone therefore does not explain the null: at the divergence level where the theory predicts the largest gain, the effect is absent, and where responses are least divergent it is adverse.

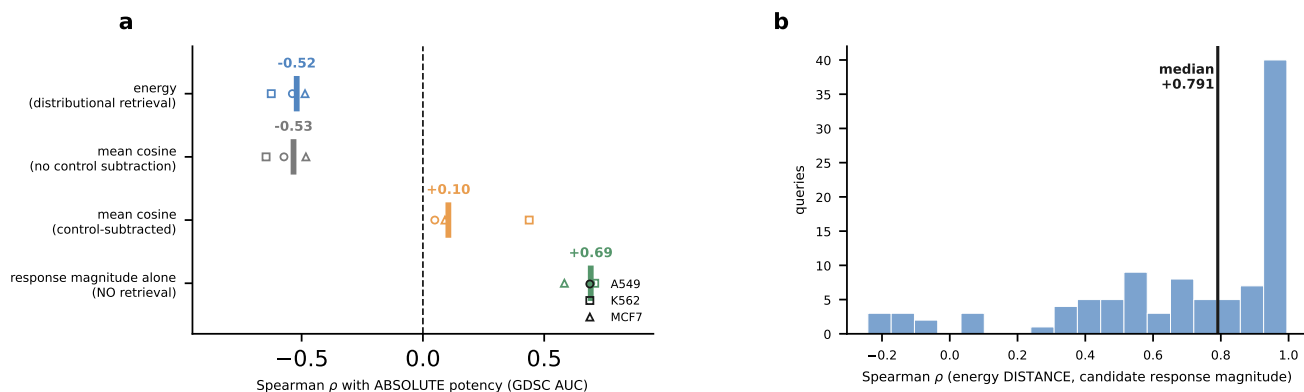


**Extended Data Fig. 3. Subpopulation identifiability in the constructed mixture, and why this panel alone could not settle it.** **a**, The original nine-column clustering sweep on a known bimodal mixture, reported here for completeness and superseded by main-text Fig. 6e. The best value across methods is a median adjusted Rand index of 0.106, an *upper bound* rather than a typical value: it is the best-case column taken over the minority of seeds in which that column wins, while the dense per-method columns lie at 0.037–0.078 and the global maximum over all methods and seeds is 0.149. The nine columns comprise only 6–7 *distinct* algorithms (the response-space  $k=2$  column is bit-identical to its raw-expression counterpart because  $k$ -means is translation-invariant, and the three best- $k$  columns are one method under three names), and *all are centroid- or Gaussian-based*. **This panel alone cannot support the conclusion once drawn from it.** Clustering failure is consistent both with the information being absent and with its being present but unreachable without labels, and those have opposite implications. Main-text Fig. 6e resolves it by adding the graph- and density-based methods missing here (Leiden, HDBSCAN) and, decisively, a supervised ceiling: the ceiling is 0.692 accuracy and the best unsupervised method reaches 0.674, so *within this constructed mixture* the limit is informational rather than algorithmic. **That qualifier turned out to be everything, and even the corrected reading was corrected again.** The 0.692 ceiling is largely an artefact of *pooling* several HDAC and several JAK compounds into two classes: split the same cells one drug against one drug and the ceiling rises to 0.879. Posed as the same drug-response question in a patient’s tumour, the ceiling is 0.923 while unsupervised clustering reaches only 0.777, a gap of +0.117 (main-text Fig. 5d): the information is present and the clustering step cannot reach it, so in real tissue the limit is *algorithmic*, not informational. An intermediate version of this correction compared the drug-response partition here with a *cell-type* partition in the tumour (malignant versus myeloid control cells, ceiling 0.964), which is a far easier question and licensed nothing; that number is withdrawn too. The general claim once drawn from this panel is withdrawn. **b**, Silhouette scores near zero across real drugs in both datasets (median 0.034; 0.4% of drugs exceed 0.1), consistent with the constructed-mixture reading. **c**, Structure diagnostics for predicted versus real candidate populations. Predicted populations *retain* baseline structure (subpopulation-variance ratio 0.142 versus 0.138 real); what they appear to lack is differential response (induced response cosine 0.19–0.37 versus 0.014 real). **That comparison does not survive scrutiny and is retained only for the structure-preservation point, which does not depend on it:** these candidate populations blend two cell lines, so an additive predictor emits a different delta per context, and for a latent model the response measured against real control cells is contaminated by the generator’s reconstruction bias. Measured within one context and against the model’s own baseline, an additive predictor returns exactly 1.000000 (main text; CORRECTIONS.md R28, R30). Predicted populations are synthesized by applying the predicted effect to the query context’s real control cells; synthesizing them instead as an isotropic Gaussian around a single mean would make every structure diagnostic a measurement of the synthesizer (isotropy index 0.998, variance ratio 0.009 by construction), see Methods. **d**, Identifiability phase diagram over separation  $\times$  cell budget: only source separation lifts ARI, and the cell-budget axis is flat, so the failure is not a sample-size problem. The separation manipulation is fit once on the full source populations, independently of the sample being evaluated; fitting it on the sampled cells and then shifting those same cells amplifies whatever separation each small draw happens to contain and yields the indefensible result that ARI *falls* as the cell budget rises. See Methods.

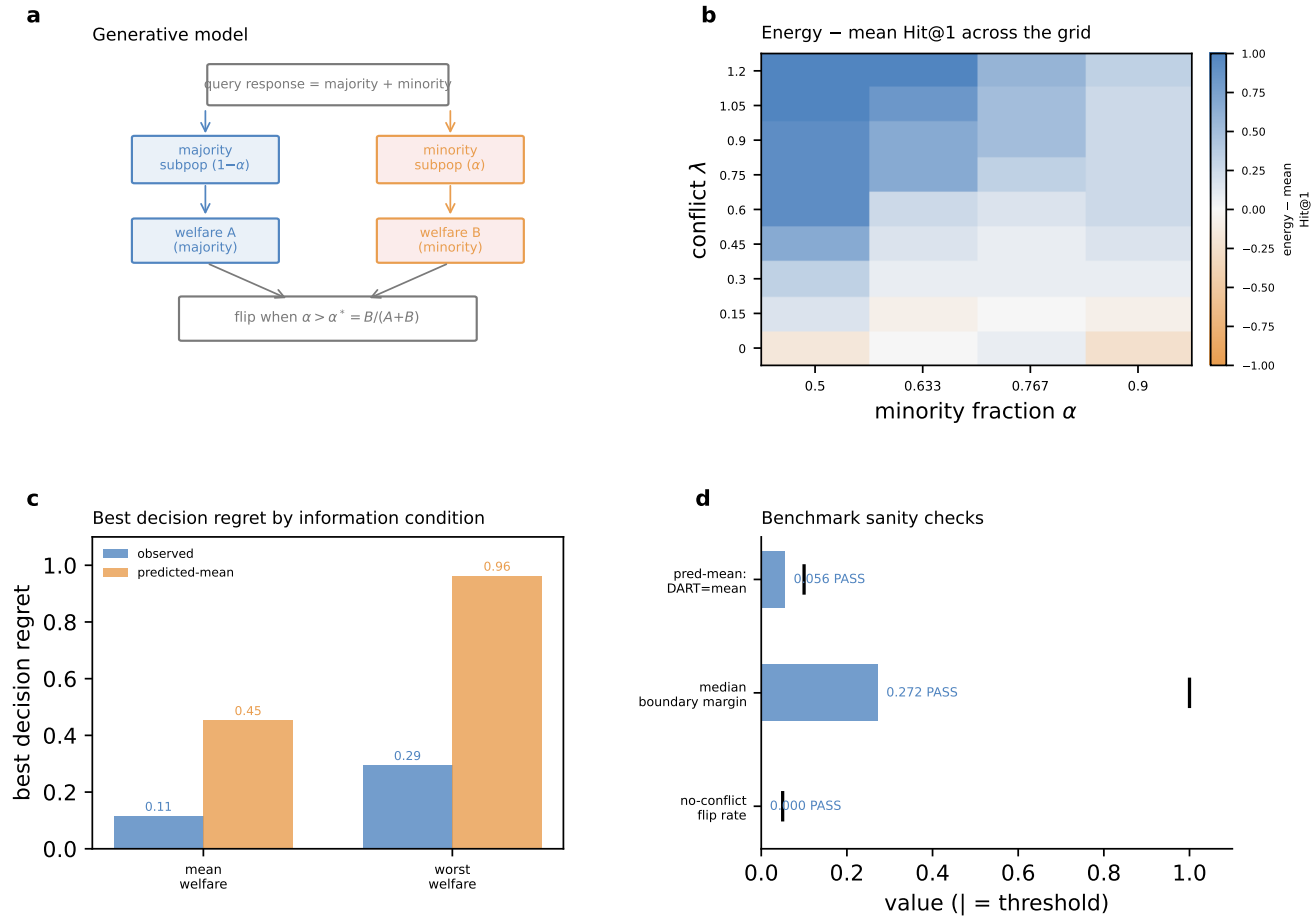


Exploratory (not claimed as robust); all signals carry qualifiers; no survival or timecourse validation

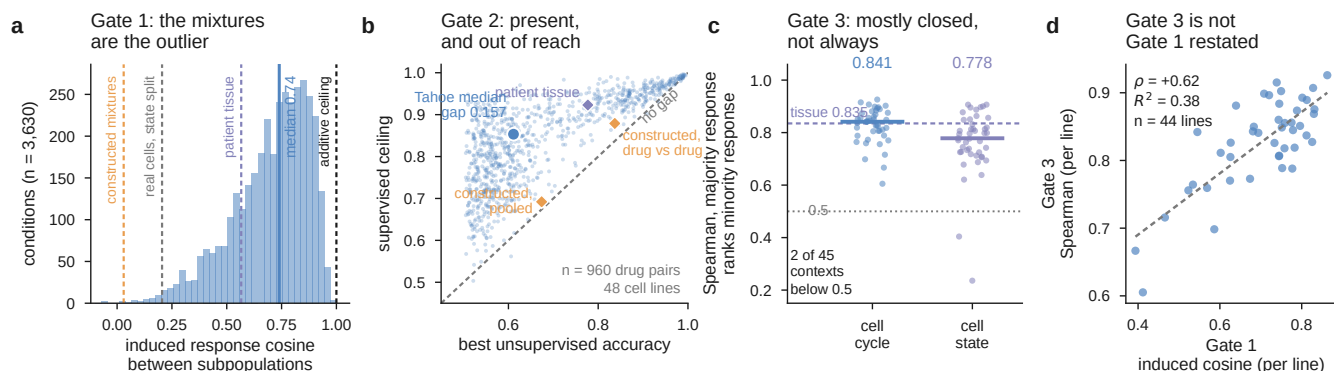
**Extended Data Fig. 4. Exploratory resistance-associated divergence.** **a**, Program enrichment in high- versus low-divergence minority states (IFN response  $p = 1.9 \times 10^{-6}$ , AXL/mesenchymal  $p = 3 \times 10^{-3}$ ). **The largest-magnitude effect in this panel is neither of those, and we do not interpret it.** The quiescence program carries an enrichment of  $-0.072$ , nearly twice the size of the next largest ( $+0.039$  for IFN response), and it is highly significant on its face ( $p = 7.7 \times 10^{-7}$ , marked \*\*\*). Its *sign* is not stable across runs, so we report the bar rather than suppress it, and we draw no conclusion from it (Supplementary Note 3). Removing it because it is inconvenient would be the selection this paper exists to criticize; leaving it unlabelled would be worse. **b**, AXL/mesenchymal enrichment versus response divergence (Pearson  $r = 0.134$ ). **c**, Minority-rescue contrast between the distributional and mean picks is near null. All signals are exploratory and hypothesis-generating, with no survival or timecourse validation.



**Extended Data Fig. 5. The absolute-potency confounder audit, and why it is not a Class-C result.** An earlier version of this work graded the retrieval rankings against *absolute* drug potency and concluded that retrieval similarity is anti-aligned with therapeutic utility. That conclusion is withdrawn, and this figure records both the analysis and the reason. **a**, Scored against absolute GDSC2 AUC, every retrieval ranking inverts (energy  $\rho = -0.52$ ; a mean baseline without control subtraction,  $-0.53$ ; the correctly specified incumbent,  $+0.10$ ) while a query-independent scalar that ranks candidates by the magnitude of their own response and never looks at the query reaches  $+0.69$ . *The oracle is mismatched to the task*. A similarity retriever is asked which candidate resembles the query, not which kills hardest; handed a weak query it should correctly return other weak drugs, and this scoring counts that as failure. Because potency is largely driven by response magnitude, a magnitude scalar is near-guaranteed to win here. No therapeutic conclusion follows in either direction. The semantically matched oracle, drug–drug functional similarity, is in main-text Fig. 4h,i. **b**, What the audit does establish, and it is the reason the magnitude control is mandatory: the energy *distance* between a query and a candidate is largely a function of the candidate’s own response magnitude (median  $\rho = +0.791$  across queries), because large-response populations sit far from everything. Ranking nearest-first therefore systematically returns the smallest-response candidates. Any distributional retrieval method claiming a functional validation must be shown to beat a scalar magnitude baseline.



**Extended Data Fig. 6. HIR-Bench: the panels demoted from the main text.** Main-text Fig. 5 keeps only the two HIR-Bench panels that carry argument, the analytic condition under which subpopulation structure can change a decision and the  $2 \times 2$  that *measures* circularity. The four panels here were previously in the main text and should not have been: HIR-Bench’s phase boundary is designed into its generator, so recovering it verifies that the benchmark works rather than establishing anything about biology, and its transfer to real data fails (Supplementary Note 1). A reader could reasonably have read four main-text panels of a synthetic benchmark as biological evidence. They are retained because they document that the benchmark behaves as specified, which is a precondition for trusting the two panels that stayed. **a**, Generative model: a query response is a majority plus a minority subpopulation with welfares A and B. **b**, Energy-minus-mean Hit@1 across the (minority fraction, conflict) grid. **c**, Best decision regret by information condition. Collapsing candidates to their means inflates regret against both objectives and most severely against the worst-welfare one (0.29 observed versus 0.96 predicted-mean), which is the objective that depends on the minority subpopulation. **d**, Benchmark sanity checks: the no-conflict flip rate is 0.000 and the median boundary margin 0.27, both within threshold.



**Extended Data Fig. 7. The three conditions re-measured on unconstructed material.** Tahoe-100M plate 3 (Zhang *et al.* 2025; 10.1101/2025.02.20.639398): a complete grid of 50 cancer cell lines against 93 compounds at one dose, a vehicle arm in every line, 4,158,278 cells, median 648 cells per condition. Its within-line heterogeneity is intrinsic cell state, not two populations combined by us, which is the property that makes it useful here. **It is not tissue, and it is used for sample size rather than for relevance;** the paper's claims about what matters therapeutically continue to rest on the patient data of main-text Fig. 5. The statistics are the paper's own, not new ones. **a**, Gate 1. Induced response cosine between the two subpopulations of a treated population, each referred to its own matched control, over 3,630 conditions (median 0.739). Dashed lines mark the four values the main text quotes. Our constructed mixtures are not at one end of the unconstructed range: only 0.11% of conditions are as divergent as they are. **b**, Gate 2, posed as the same drug-response question and scored with the same four-method clusterer panel. One point per drug pair; diamonds are the three main-text settings. Supervised ceiling 0.853 against best unsupervised 0.611, a gap of 0.157 that is positive in 48 of 48 cell lines. No method family rescues it (k-means 0.592, Gaussian mixtures 0.582, Leiden 0.563), which differs from the tumour, where Leiden was the method that partly survived. **c**, Gate 3 under two partitions, both computed on *disjoint* cell sets, against the tumour value computed the same way (0.835). Medians 0.841 and 0.778. The condition is not universal: 2 of 45 contexts fall below 0.5 under the control-state partition, the lowest at 0.236, and those are the contexts in which a distributional score should pay off. **d**, Gate 3 against Gate 1 within the cell-cycle partition, which is the only comparison of the two that is like-for-like. They share some variance and are not the same condition. **Limits.** These are cell lines; this is one plate of fourteen and 93 of 380 compounds; both partitions are coarse two-way splits of cell state rather than the persister axis that motivates the paper; and which contexts show an open Gate 3 depends on the partition, the two agreeing at the extremes but only moderately in rank (Spearman 0.466).

6 **Supplementary Note 1. HIR-Bench: sanity checks, and why the transfer test's 100% flag rate**  
7 **is not evidence.**

8 The benchmark's internal checks hold on the full grid: the no-conflict flip rate is 0.0000, as it must be for  
9 a generator whose flip boundary is  $\alpha^* = B/(A + B)$ , and the median boundary margin is 0.27.

10 Under the information-condition transfer test, the gate flags mean-collapsed predicted responses as  
11 no-distributional in 100% of cases. We decline to count that number as evidence, because it is a **tautology**:  
12 the rule returns no-distributional *unconditionally* whenever the information condition is predicted-mean.  
13 It cannot return anything else, so the 100% restates the rule rather than testing it. The substantive claim,  
14 that mean-collapsed candidate libraries offer no distributional advantage, rests on the predict-then-rank  
15 experiments of the main text, not on this figure.

16 The projection of the HIR-Bench regime boundary onto real data **fails its own acceptance test, and**  
17 **we report it as failing.** The label is heavily imbalanced (228 to 11), so a constant predictor scores 95.4%;  
18 the regime projection scores 61.1%, which is 34 points *worse* than saying nothing. The boundary does  
19 not transfer. Two structural limits bound the test in both directions and we work around neither: the  
20 boundary is fitted on the method-performance layer, which is available on a 144-instance grid (72 grid  
21 cells) because scoring nine retrieval methods against up to 100 candidates per instance does not scale to  
22 the full grid; and the acceptance label derives from the same uncalibrated +0.01 threshold the main text  
23 declines to trust.

24 **Supplementary Note 2. Two further tests of the two-gate criterion.**

25 **(a) Coverage performance versus query identifiability.** The two gates predict where a subpopulation-  
26 coverage score should and should not carry signal. Stratifying real-data coverage by query identifiability  
27 is consistent with them, but weakly and *not* monotonically. Coverage MoA-nDCG is higher for more  
28 identifiable queries (Spearman  $\rho = 0.17$ ,  $p = 6.3 \times 10^{-5}$ ,  $n = 543$  query-seed pairs), but the lift is  
29 confined to the top tertile: tertile means are 0.285, 0.240 and 0.444 from least to most identifiable, so the  
30 middle tertile sits *below* the least-identifiable one. Two caveats bound even that. The most-identifiable  
31 tertile carries a median of 6 same-mechanism candidates against 2 in the other two tertiles, and a larger  
32 positive pool mechanically raises nDCG, so the correlation is not cleanly separated from candidate-pool  
33 size. And Gate 2 is almost never open *in these SciPlex3 queries*: even the most-identifiable tertile has  
34 a median silhouette of 0.038. That scope matters. On patient glioblastoma the same gate is wide open  
35 (main-text Fig. 5d), so the statement here is a fact about the cell-line data this note analyses and not about  
36 biology.

37 **(b) The conditional advantage the criterion predicts is absent.** If identifiable structure is what lets  
38 distributional scores help, the per-query mechanism-recovery advantage should turn positive where  
39 structure is present. Stratifying 600 leave-drug-out queries by response divergence, it does rise, and the  
40 ordering is real and leakage-free (Spearman  $\rho = 0.14$ , permutation  $p = 2 \times 10^{-4}$ ). But it never clears  
41 parity: the most-divergent stratum reaches only +0.02 in nDCG, with a bootstrap interval spanning  
42 zero, and it reverses in one of three cell lines. Decisively, when queries are stratified by the actual Gate-2  
43 variable, query identifiability (silhouette), the advantage is flat ( $\rho \approx 0$ ). Response divergence measures  
44 response spread, not resolvable subpopulation structure, and Gate 2 does not open *in these SciPlex3*  
45 *queries* even for the most response-divergent ones. **This is not a statement about real tissue**, and an  
46 earlier version of this note wrote it as though it were. On patient glioblastoma Gate 2 opens (main-text  
47 Fig. 5d) and the distributional advantage still does not follow, which is why the two-gate criterion is  
48 reported as necessary but not sufficient.

49 **Supplementary Note 3. Exploratory resistance-associated divergence.**



50 In the Frangieh melanoma Perturb-CITE-seq data, cells whose response direction diverges most between  
51 immune contexts are enriched for resistance-associated programs: an AXL/mesenchymal program (cor-  
52 relation +0.134,  $p = 0.003$ ), an interferon-response program ( $p = 1.9 \times 10^{-6}$ ), and antigen-presentation  
53 genes ( $p = 0.04$ ); see Extended Data Fig. 4.

54 This analysis is **exploratory and hypothesis-generating throughout** and is reported as a direction for  
55 future work, not as a result. There is no longitudinal drug timecourse, no survival readout, and no post-  
56 treatment clone measurement, so we make **no claim that a distributional score predicts resistance** or  
57 identifies persister cells. A quiescence program's sign was not stable across runs and is not interpreted.  
58 Consistent with the rest of the study, the distributional-versus-mean minority-rescue contrast is itself  
59 near-null. The observation motivates future experiments with functional readouts, and nothing more.

Supplementary Table 1. Datasets, preprocessing, and query construction.

(a) **Dataset scale.** Four benchmarks span controlled proof, natural heterogeneity, an honest negative control, and a synthetic analytic stress test.

| Dataset                         | Cells                        | Genes     | Context axis                        | Perturbations           |
|---------------------------------|------------------------------|-----------|-------------------------------------|-------------------------|
| SciPlex3 (Srivatsan 2020)       | 276,325                      | 2,000 HVG | A549, K562, MCF7                    | 188 drugs (all 3 lines) |
| Frangieh (2021)                 | 218,023                      | 2,000 HVG | Control, IFN $\gamma$ , Co-culture  | 239 CRISPR KOs          |
| CD34+ (GSE306429)               | 33,984                       | 2,000 HVG | discovered subpopulations           | 36 drugs                |
| HIR-Bench (synthetic instances) | 13,440                       | latent    | two-subpopulation welfare model     | tunable conflict        |
| GDSC2 dose-response (Class C)   | 286 drugs $\times$ 969 lines |           | 966 lines (SciPlex3 lines held out) | 35 matched compounds    |
| Tahoe-100M plate 3 (Zhang 2025) | 4,158,278                    | 2,304 HVG | 50 cancer cell lines                | 93 compounds, 1 dose    |

(b) **Preprocessing.** Processed tensors are built once and read directly by the retrieval code; the build scripts are kept for provenance.

| Step           | Setting                                                                                        |
|----------------|------------------------------------------------------------------------------------------------|
| Source         | SciPlex3: Srivatsan & Trapnell 2020; Frangieh: scPerturb Zenodo; CD34+: GSE306429 counts layer |
| Normalization  | log-normalized expression; 2,000 highly variable genes selected per dataset                    |
| Controls       | real DMSO controls (SciPlex3); condition controls (Frangieh); DMSO (CD34+)                     |
| Response delta | per-cell expression minus matched control centroid, used only where a metric requires it       |
| Dose           | 10,000 nM (SciPlex3 controlled and cross-line experiments)                                     |

(c) **Query construction.** A query is a mixture of two response modes at mixing weight  $\alpha$ ; the retrieval task is to rank the candidate that generated it above distractors.

| Parameter              | Controlled (SciPlex3)           | Cross-line / natural                   |
|------------------------|---------------------------------|----------------------------------------|
| Response modes         | HDAC (class a) vs JAK (class b) | cell-line pairs / immune-context pairs |
| Query size             | 400 cells                       | 200 cells                              |
| Distractors            | 40                              | 40                                     |
| Mixing weight $\alpha$ | 0.5, 0.6, 0.7, 0.8, 0.9         | 0.5–0.9                                |
| Seeds                  | 20 (base 1000)                  | 10 (cross-line), 20 (Frangieh)         |
| Identifiability probe  | min 30 cells, gate 0.9          | min 30–40 cells, gate 0.9              |

Cell and gene counts are the processed-tensor totals. HIR-Bench’s 13,440 entries are synthetic benchmark *instances* (672 parameter configurations  $\times$  20 seeds), not biological cells; they are the grid used for the predictability analysis (Fig. 6b).

69 **Supplementary Table 2. Field-level evaluation audit.**

70 The claim that objective-aligned evaluation is a field-wide default should be checkable rather than as-  
71 serted, so the audit is reproduced here rather than left with the code. **Inclusion criteria:** a method family  
72 qualifies if (i) it ranks or selects perturbations, or predicts perturbation responses that a ranking layer  
73 consumes, (ii) it is represented by at least one widely used published implementation, and (iii) its head-  
74 line evaluation metric is stated in the source. **Alignment** is judged *high* when the headline metric is a  
75 monotone transform of the scoring rule’s own objective evaluated on the same data, which is the condi-  
76 tion under which a gain is close to guaranteed. Families are the unit; we do not score individual papers,  
77 and this is an audit of *evaluation practice*, not of correctness.

| Method family                                 | Representative meth-<br>ods                                                                                      | Objective                                                      | Headline evaluation                                                                                  | Metric<br>class         | Alignment risk                                                                  |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------|
| CMap / L1000<br>signature re-<br>trieval      | Connectivity Map (Lamb <i>et al.</i> 2006); L1000 / WTCS connectivity (Subramanian <i>et al.</i> 2017)           | rank by connectivity of mean differential signatures           | connectivity score; cosine of mean signatures                                                        | A                       | high: the metric is the scoring rule restated                                   |
| Single-cell signature retrieval               | pseudobulk and single-cell connectivity on harmonized perturbation atlases (Peidli <i>et al.</i> 2024)           | rank by single-cell-derived mean or pseudobulk signature match | signature correlation; Hit@k; DEG overlap                                                            | A (B via DEG)           | high: signature-space metric mirrors signature-space scoring                    |
| Distributional retrieval (this work)          | energy, MMD, sliced-Wasserstein, subpopulation coverage                                                          | rank by population-to-population distributional distance       | energy/MMD welfare-regret proxy (A); MoA-nDCG, minority coverage (B); GDSC functional similarity (C) | A + B + C               | high on the Class-A proxy; self-raised and measured here                        |
| Single-cell perturbation predictors           | scGen (Lotfollahi <i>et al.</i> 2019); CPA (Lotfollahi <i>et al.</i> 2023); chemCPA (Hetzl <i>et al.</i> 2022)   | predict post-perturbation expression from control              | reconstruction MSE / $R^2$ / PCC of predicted delta                                                  | A (B via DEG)           | high: training loss and evaluation share the reconstruction target              |
| Virtual-cell / perturbation foundation models | scGPT (Cui <i>et al.</i> 2024); scFoundation (Hao <i>et al.</i> 2024); Geneformer (Theodoris <i>et al.</i> 2023) | general-purpose prediction of perturbed cell state             | held-out reconstruction; embedding similarity                                                        | A + B                   | high: reconstruction and embedding metrics align with the pretraining objective |
| Graph / target-based inverse design           | GEARS (Roohani <i>et al.</i> 2024); network-proximity repositioning (Guney <i>et al.</i> 2016)                   | rank by graph proximity or target overlap                      | graph proximity; target-family overlap                                                               | A                       | high: the scoring graph and the evaluation graph coincide                       |
| Target/drug graph methods (PDGrapher-style)   | PDGrapher (Gonzalez <i>et al.</i> 2026)                                                                          | predict interventions moving a cell state toward a target      | closed-loop target ranking recall/nDCG                                                               | A (B via target family) | high: the closed-loop metric is aligned with the graph objective                |

78 The table audits **six published method families**, with the present work listed as a seventh row for contrast and excluded from  
79 every count below. Each of the six published families has a headline metric in Class A, and none reports a Class-C readout as

80 its primary evidence. This is the pattern the present study raises against itself: we report all three classes on one task set, and  
81 the Class-A gain does not survive the other two at anything near its apparent size. The unit of the audit is the *family*, not the  
82 paper, and it is an audit of evaluation practice rather than of correctness; we do not score individual papers. The **representative**  
83 **methods** column names widely used implementations so that each row can be checked against specific work rather than taken  
84 on assertion; being named there is not a criticism of a method, and the objective and headline-evaluation columns describe the  
85 family's common practice rather than any one paper's full evaluation. Full citations with DOIs accompany the code.

86 **Supplementary Table 3. Cell-compartment validation in the natural-heterogeneity bench-**  
87 **mark.**

88 The natural-heterogeneity result (main-text Fig. 5d–f) depends entirely on the cell-type labels being what  
89 they claim to be, so the labelling is validated rather than trusted. Each cell receives a z-scored marker  
90 score per compartment and is called only if it scores highest for that compartment, clears an absolute  
91 expression floor on that compartment’s key markers, and beats the runner-up by a margin; everything  
92 else is labelled unassigned and *dropped* (39.9% of cells). The table below reports, for each called com-  
93 partment, the mean log-normalized expression of every compartment’s key marker set. A compartment  
94 passes only if its own markers are the most strongly expressed. The loader raises rather than writing  
95 labels if any compartment fails.

| Called compartment    | Cells  | Mean expression of key markers for |              |                 | Passes |
|-----------------------|--------|------------------------------------|--------------|-----------------|--------|
|                       |        | malignant                          | myeloid      | oligodendrocyte |        |
| Malignant glioma      | 41,314 | <b>1.024</b>                       | 0.073        | 0.117           | yes    |
| Tumour-assoc. myeloid | 31,678 | 0.121                              | <b>1.534</b> | 0.169           | yes    |
| Oligodendrocyte       | 23,233 | 0.409                              | 0.114        | <b>1.784</b>    | yes    |

96 **What this table exists to prevent.** A first version of the assignment took a bare argmax over five marker sets, which forces  
97 every cell into some compartment, including cells expressing none of the markers. It produced two compartments that fail  
98 exactly this check: cells labelled T cells (22% of the dataset) had a mean CD3D/CD3E/CD2 expression of 0.080, *lower* than  
99 their malignant-marker score of 0.141, and cells labelled endothelial expressed myeloid markers (1.074) more strongly than  
100 endothelial ones (0.951). They were neither. Building a natural-heterogeneity result on a fabricated T-cell compartment would  
101 have been worse than reporting no natural dataset at all, and only the three compartments above are used.

102 **Supplementary Table 4. The Class-C functional oracle, under both constructions.**

103 Cell lines differ greatly in general drug sensitivity, and that shared main effect inflates every drug–drug  
 104 correlation: left uncorrected, 85% of drug pairs appear positively similar (median +0.178). The oracle  
 105 used in the main text therefore subtracts each cell line’s mean AUC, estimated across all 286 GDSC2 com-  
 106 pounds, before correlating; the corrected oracle is symmetric about zero (52% positive, median +0.015).  
 107 **That choice decides the result, and the table below is why we state the dependence in the main text**  
 108 **rather than reporting only the ordering we prefer.** Values are median per-query Spearman  $\rho$  against  
 109 the oracle, over 103 leave-one-drug-out queries in three cell lines.

| Ranking rule                                     | Corrected oracle<br>(cell-line mean removed) | Uncorrected oracle<br>(raw AUC profiles) |
|--------------------------------------------------|----------------------------------------------|------------------------------------------|
| Energy distance (the distributional score)       | <b>+0.276</b>                                | +0.097                                   |
| Mean cosine, control-subtracted (the incumbent)  | +0.083                                       | <b>+0.138</b>                            |
| Mean cosine, no control subtraction              | +0.241                                       | +0.092                                   |
| Magnitude matching (query-dependent scalar)      | +0.232                                       | <b>+0.217</b>                            |
| Magnitude only (query-independent)               | −0.330                                       | +0.135                                   |
| Potency matching (diagnostic, not a recommender) | +0.399                                       | +0.290                                   |
| Energy, partialled on magnitude matching         | +0.097                                       | n/a                                      |

110 **What flips and what does not.** Under the corrected oracle the distributional score leads its incumbent threefold (+0.276 against  
 111 +0.083), and that is the one criterion in this study, invisible to every scorer, on which it wins. Under the uncorrected oracle it  
 112 does not win at all: energy falls to +0.097, *below* the incumbent’s +0.138 and well below the magnitude scalar’s +0.217. The  
 113 affirmative Class-C finding is therefore conditional on the centering. We think the centering is right, we specified it on the  
 114 reasoning above before inspecting any retrieval result, and leaving a shared cell-line main effect inside an oracle would be an  
 115 error in its own right. But a paper whose thesis is that the choice of criterion decides the winner cannot quietly exempt its own  
 116 criterion, and an earlier draft of this work asserted that both oracles “give the same verdict”, which the table shows is false.  
 117 **What survives both constructions is the negative result:** under neither oracle does the distributional score significantly beat  
 118 a scalar that compares no distributions at all. Note also that energy’s uncorrected value (+0.097) and its magnitude-partialled  
 119 value under the corrected oracle (+0.097) coincide to three decimals by arithmetic accident; they are different quantities (0.0967  
 120 and 0.0969 before rounding).

## 121 **Supplementary Note 4. What this study does not claim.**

122 This note states the scope of every claim in the main text. It is reproduced in full from an earlier main-text  
123 box and is referenced from the Discussion.

- 124 1. **We do not claim that the predictors tested here represent the field.** The in-repository linear-latent  
125 model (PCA encoder, additive latent delta, linear decoder) is additive *by construction*, so its Gate-1  
126 value demonstrates an algebraic property rather than surveying deployed models. We *do* now report  
127 one deployed model: the published scGen variational autoencoder passes its learning check and  
128 returns a true induced divergence of 0.972 against an additive ceiling of 1.000, so it is near-additive  
129 in practice. That is one model, not a field.
- 130 2. **We show that a non-additive predictor opens Gate 1, but not that opening it buys retrieval gain,  
131 and the second half is the experiment that would falsify us.** A latent-space entropic optimal-  
132 transport map passes its learning check (0.352) and reaches a true induced divergence of 0.900, sig-  
133 nificantly below the additive ceiling of 1, so Gate 1 does open; but only a crack, since real cells sit at  
134 0.205, and the map’s learning is weak. The same map run naively in gene space did not clear the bar  
135 (learning 0.057), and the published CPA model, though it reconstructs cells at  $R^2 \approx 0.39$ , does not  
136 learn per-drug effects at either sixty or three hundred cells per drug (delta learning 0.08 in both) and  
137 is therefore void. We did re-run the predict-then-rank retrieval with this Gate-1-opening map, and  
138 it was inconclusive: the OT map’s cross-line predictions are too poor to retrieve with at all (Hit@1  
139 0.05 under every rule, against 0.88 for the additive predictor), so we lack a predictor that both pre-  
140 dictors accurately and opens Gate 1, and whether opening Gate 1 buys a distributional advantage stays  
141 untested. A modern flow-matching model, CellFlow, conditioned on ECFP4 compound fingerprints,  
142 learns within context (mean-delta cosine 0.24, opening Gate 1 at induced divergence 0.85) but col-  
143 lapses to 0.03 under leave-one-context-out at a matched budget, an  $8.4\times$  drop; per drug this transfer  
144 tracks cross-context response similarity ( $r = 0.85$ ), so it too cannot serve as the clean cross-context  
145 predictor the falsification requires. The collapse holds under distributional (energy) and differential-  
146 expression metrics, not only the mean-delta check; and conditioning CPA on the same ECFP finger-  
147 prints instead of a learned embedding does not rescue it (learning 0.03), locating that failure in the  
148 adversarial objective rather than the drug representation.
- 149 3. **We do not claim that the distributional score provides better therapeutic recommendations.** Every  
150 gain in this study is measured under an objective-aligned (Class A), task-proximal (Class B), or semi-  
151 real functional (Class C) criterion. None is a clinical or same-cell endpoint.
- 152 4. **We do not claim that the functional-oracle advantage is a distributional effect, nor that it is ro-  
153 bust to how the oracle is built.** Energy retrieval does beat the correctly specified mean incumbent  
154 under the cell-line-centered GDSC oracle (+0.276 versus +0.083), but a query-dependent scalar that  
155 compares no distributions reaches +0.232, energy beats it on only 62 of 103 queries ( $p = 0.073$ , it-  
156 self anticonservative), and partialling that magnitude channel out leaves +0.097: the gain is real and  
157 mostly not distributional. It is also conditional on the centering, since under the uncorrected oracle  
158 energy falls to +0.097 and loses to both the mean incumbent (+0.138) and the scalar (Supplementary  
159 Table 4). We specified the centering in advance and argue it is correct, but only the negative survives  
160 both constructions: under neither oracle does the distributional score significantly beat a scalar that  
161 compares no distributions.
- 162 5. **We do not claim that retrieval similarity is anti-aligned with therapeutic utility.** An earlier framing  
163 of this work graded the rankings against *absolute potency* and drew that conclusion. It does not follow:  
164 a similarity retriever handed a weak query should return other weak drugs, and potency is largely  
165 driven by response magnitude, so a magnitude scalar is near-guaranteed to win that comparison. The  
166 absolute-potency analysis is retained only as a confounder audit exposing the magnitude channel.
- 167 6. **We do not claim an improvement under the task-proximal metric.** On the same scorer and the

same queries that earn a Class-A gain of +0.129 (median), the mechanism-recovery gain is 0.000 (median) and  $-0.037$  (mean). Across 239 real-data tasks the coverage advantage is statistically real and negligible (mean +0.0018) and lives only in constructed cell-line mixtures.

7. **We do not report a count of “distributionally dominant” real tasks, and no such count should be quoted from this work.** The conventional threshold sits inside the noise band of the difference distribution, and the seed re-draws which drugs are tested and re-clusters the minority subpopulation, so it is not a replicate. Under resampling only 2 of the 10 threshold-crossing cross-line tasks survive, while drugs that cross the threshold at no seed in the recorded run cross it in 1 to 4 of 10 resamples.

8. **We do not claim that HDAC inhibitors are a validated regime for distributional retrieval.** Exactly two tasks survive seed resampling, both HDAC inhibitors in one cell-line pair. With  $n = 2$  that is a hypothesis, not a result.

9. **We do not claim that minority-state coverage is oracle-independent.** It is a mean-based cosine proxy and is not the distributional score’s own objective, but it rewards covering the minority state, which is what the worst-case coverage scorer optimizes. It is task-proximal, not independent. Of the judges used here, MoA recovery is annotation-derived and the GDSC functional oracle is the only externally grounded one; the distributional score is worse than the incumbent on the first and better on the second, by a margin a scalar reproduces.

10. **We do not claim that the current gate reliably transfers to real retrieval tasks.** The gate is experimental; it does not isolate oracle-independent gains in partial-observed real data, its reliability axis is anti-correlated with true divergence ( $\rho = -0.21$ ), and the dataset-level divergence criterion admits our negative control (CD34+, cosine 0.186) just as readily as our positive anchor. Its apparent success at flagging mean-collapsed predicted responses is a tautology: the rule returns no-distributional unconditionally for that information condition.

11. **We do not claim that HIR-Bench is a biological ground truth.** It is a constructive synthetic benchmark whose phase boundary is expected by design.

12. **We do not claim that subpopulation identifiability is an information limit, and we do not claim the opposite either.** Both readings appeared in earlier drafts and both are withdrawn. The first came from a constructed mixture that pooled several drugs into each class, which lowered its ceiling for reasons unrelated to biology. The second came from comparing that drug-response partition with a *cell-type* partition in a tumour, which is an easier question and returned a reassuring 0.964 that meant nothing. Posed as the same question in both settings, the tumour ceiling is 0.923 and unsupervised clustering reaches 0.777: the information is present and the clustering step cannot reach it, so in real tissue the limit is algorithmic. That result rests on 4 patients, 30 of its 36 splits from one of them, and a compartment assignment that discards 39.9% of cells, and it should be treated accordingly.

13. **We do not claim that our structure-preservation and induced-divergence diagnostics measure the perturbation rather than the generator.** A latent predictor returns  $P(C) + W\delta_z$ , the autoencoder’s reconstruction plus the decoded delta. Referred to its own decoded baseline, an additive latent model induces *exactly zero* response divergence ( $\cos = 1.000000$ ). Referred to the *real* control cells, which is what any retrieval scorer does, the same model shows apparent divergence as low as  $-0.13$ , and every bit of it is reconstruction bias. Our subpopulation-variance-ratio and induced-cosine diagnostics are computed against real controls and therefore inherit this contamination. We report both baselines and flag the affected numbers rather than quietly choosing the flattering one.

14. **We do not claim that the distributional score is better on the surface-protein oracle, nor that the mean is.** The winner is determined by whether that oracle is built as a mean or as a distribution, from the same proteins in the same cells. We report the reversal, not a victor. What the reversal does establish is that objective alignment is not confined to a method grading its own objective: it operates between external, blind criteria, and it therefore threatens external validations too, including ours.



- 215 15. **We do not claim that the distributional score predicts resistance.** The resistance analysis is ex-  
216 ploratory and hypothesis-generating; there is no timecourse or survival readout, and the minority-  
217 rescue contrast is near-null.
- 218 16. **We do not benchmark live PDGrapher, and no number in this paper derives from LINCS.** Pertur-  
219 bation predictors are treated as candidate-response sources, not as models under test.
- 220 17. **We do not claim that all CMap implementations are algebraically identical to the score imple-**  
221 **mented here;** the exact identity is limited to the implemented cosine mean-signature kernel.